Differentiating sin, cos, tan, e^x and $\ln x$

Six trigonometric functions, the exponential and the logarithm — where each derivative comes from, and the three rules that extend them to everything else.

Pure Mathematics 1, 7.3–7.4 · Pure Mathematics 2, 4.1–4.4 · explanations and 30 questions

KEY WORDS

- radian
- standard limit
- first principles
- chain rule
- product rule
- quotient rule
- natural logarithm
- exponential function
- logarithmic differentiation

REMEMBER

1. $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$ — the minus sign on the cosine is the mark most often lost.
2. $(e^x)' = e^x$: the one function that is its own derivative. $(\ln x)' = \dfrac{1}{x}$.
3. Radians, always. In degrees $(\sin x°)' = \dfrac{\pi}{180}\cos x°$ instead.
4. Chain rule: differentiate the outside, then multiply by the derivative of the inside.
5. $\dfrac{d}{dx}\ln f(x) = \dfrac{f'(x)}{f(x)}$ — worth knowing on sight.

Why radians, and not degrees

Every formula below is false in degrees. An angle of x degrees is $\dfrac{\pi x}{180}$ radians, so $\sin x° = \sin\dfrac{\pi x}{180}$, and the chain rule drags that constant through: $(\sin x°)' = \dfrac{\pi}{180}\cos x° \approx 0.01745\cos x°$.

Radians are the measure in which the constant comes out as exactly 1. That is the whole reason calculus insists on them.

The four limits everything rests on

Each derivative below is built from one of these. They are worth memorising as facts, not re-derived each time.

LIMIT AS $h \to 0$	VALUE	GIVES
$\dfrac{\sin h}{h}$	1	the derivative of $\sin x$
$\dfrac{1-\cos h}{h}$	0	the derivative of $\sin x$
$\dfrac{e^h - 1}{h}$	1	the derivative of e^x

LIMIT AS $h \to 0$	VALUE	GIVES
$\dfrac{\ln(1+h)}{h}$	1	the derivative of $\ln x$

Where $(\sin x)' = \cos x$ comes from

Start from the definition and expand with the addition formula.

1. $\dfrac{\sin(x+h) - \sin x}{h}$ the difference quotient
2. $= \dfrac{\sin x \cdot \cos h + \cos x \cdot \sin h - \sin x}{h}$ addition formula for $\sin(x+h)$
3. $= \cos x \cdot \dfrac{\sin h}{h} - \sin x \cdot \dfrac{1 - \cos h}{h}$ split, and group the two standard limits
4. $\to \cos x \cdot 1 - \sin x \cdot 0 = \cos x$ let $h \to 0$

Why e is the base that works

For any base, $(a^x)' = a^x \cdot \lim_{h\to 0} \dfrac{a^h - 1}{h}$. That limit is a constant depending on a, and it equals $\ln a$. So $(a^x)' = a^x \ln a$, and the constant is 1 — the function reproducing itself exactly — only when $a = e$.

The logarithm follows by reversing it: if $y = \ln x$ then $e^y = x$, and differentiating both sides gives $e^y \dfrac{dy}{dx} = 1$, so $\dfrac{dy}{dx} = \dfrac{1}{e^y} = \dfrac{1}{x}$.

Note $(\ln|x|)' = \dfrac{1}{x}$ as well, which is what lets $\dfrac{1}{x}$ be integrated for negative x.

The table

Every function in the right-hand column of the trigonometric half — the co-functions — carries a minus sign. That is the pattern to hang them on.

FUNCTION	DERIVATIVE	FUNCTION	DERIVATIVE
$\sin x$	$\cos x$	e^x	e^x
$\cos x$	$-\sin x$	e^{ax+b}	ae^{ax+b}
$\tan x$	$\sec^2 x = \dfrac{1}{\cos^2 x}$	a^x	$a^x \ln a$
$\cot x$	$-\operatorname{cosec}^2 x$	$\ln x$	$\dfrac{1}{x}$
$\sec x$	$\sec x \cdot \tan x$	$\ln(ax+b)$	$\dfrac{a}{ax+b}$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cdot \cot x$	$\log_a x$	$\dfrac{1}{x \ln a}$

The chain rule does the rest

Watch the difference between $\sin^2 x$, which is $(\sin x)^2$, and $\sin(x^2)$. They are different functions with different derivatives.

FUNCTION	DERIVATIVE	THE INNER DERIVATIVE
$\sin 3x$	$3\cos 3x$	the 3
e^{5x}	$5e^{5x}$	the 5
$\ln(2x + 1)$	$\dfrac{2}{2x + 1}$	the 2
$\sin(x^2)$	$2x \cdot \cos(x^2)$	the $2x$
$\sin^3 x$	$3\sin^2 x \cdot \cos x$	the $\cos x$
$e^{\sin x}$	$\cos x \cdot e^{\sin x}$	the $\cos x$

Worked example 1 — a product: $y = x^2 \ln x$

1. $u = x^2$, so $u' = 2x$ name the first part
2. $v = \ln x$, so $v' = \dfrac{1}{x}$ name the second
3. $y' = 2x \cdot \ln x + x^2 \cdot \dfrac{1}{x}$ apply $u'v + uv'$
4. $y' = 2x \ln x + x = x(2 \ln x + 1)$ simplify, then factorise

Worked example 2 — a quotient: $y = \dfrac{\sin x}{x}$

1. $u = \sin x$, so $u' = \cos x$ the numerator
2. $v = x$, so $v' = 1$ the denominator
3. $y' = \dfrac{\cos x \cdot x - \sin x \cdot 1}{x^2}$ apply $\dfrac{u'v - uv'}{v^2}$ — keep the order
4. $y' = \dfrac{x \cos x - \sin x}{x^2}$ tidy the numerator

The five mistakes to watch for

1. Working in degrees. 2. Losing the minus sign on $(\cos x)'$. 3. Reading $\sin^2 x$ as $\sin(x^2)$. 4. Forgetting the inner derivative — writing $(\sin 3x)' = \cos 3x$ instead of $3\cos 3x$. 5. Treating $\ln(x^3 + 1)$ as if the chain rule did not apply: it is $\dfrac{3x^2}{x^3 + 1}$, not $\dfrac{1}{x^3 + 1}$.

Every one of these can be caught by substituting a number into the original and into the derivative and comparing with a numerical gradient.

Easy EASY 10 questions

Straight from the table, one step at a time.

1. Differentiate $y = \sin x$.

2. Differentiate $y = \cos x$.

3. Differentiate $y = \tan x$.

4. Differentiate $y = e^x$.

5. Differentiate $y = \ln x$.

6. Differentiate $y = \sin 3x$.

7. Differentiate $y = e^{2x}$.

8. Differentiate $y = \ln 5x$.

9. Differentiate $y = 2 \sin x + 3 \cos x$.

10. Differentiate $y = 2^x$.

Medium MEDIUM 10 questions

A chain, a product or a quotient — one extra move.

1. Differentiate $y = \sin(x^2)$.

2 Differentiate $y = \sin^2 x$.

3 Differentiate $y = x \sin x$.

4 Differentiate $y = e^{\sin x}$.

5 Differentiate $y = \ln(\cos x)$.

6 Differentiate $y = \dfrac{\ln x}{x}$.

7 Differentiate $y = \dfrac{\tan x}{x}$.

8 Differentiate $y = e^x \sin x$.

9 Differentiate $y = \ln(x^2 + 1)$.

10 Differentiate $y = \tan(3x + 1)$.

Hard

HARD 10 questions

Several ideas at once, and some that need an idea rather than a method.

1 Differentiate $y = \dfrac{\sin x}{1 + \cos x}$ and simplify fully.

2 Differentiate $y = \ln(\sec x + \tan x)$.

3 Differentiate $y = \sin(\cos x)$.

4 Differentiate $y = x^x$ for $x > 0$.

5 Differentiate $y = (\sin x)^x$ where $\sin x > 0$.

6 Differentiate $y = \dfrac{e^x}{1 + e^x}$ and simplify.

7 Find the least value of $y = x \ln x$ for $x > 0$.

8 Find the second derivative of $y = e^{-x} \sin x$.

9 Find and classify the stationary points of $y = \sin x + \cos x$ on $0 \leq x \leq 2\pi$.

10 Prove that $(\cot x)' = -\operatorname{cosec}^2 x$ using the quotient rule.

Answers

Topic 1 · Differentiating sin, cos, tan, e^x and ln x

Easy · easy

1 $\cos x$

2 $-\sin x$

3 $\sec^2 x$

4 e^x

5 $\dfrac{1}{x}$

6 $3\cos 3x$

7 $2e^{2x}$

8 $\dfrac{1}{x}$

9 $2\cos x - 3\sin x$

10 $2^x \ln 2$

Medium · medium

1 $2x\cos(x^2)$

Chain rule with inner function x^2. The outer derivative is $\cos(x^2)$, the inner is $2x$, so the answer is $2x\cos(x^2)$. Note this is not $\sin^2 x$.

2 $2\sin x\cos x = \sin 2x$

Here the outer function is the square, so the derivative is $2\sin x$ times the derivative of $\sin x$, giving $2\sin x\cos x$. The double-angle identity tidies it to $\sin 2x$.

3 $\sin x + x\cos x$

A product: $u = x$ with $u' = 1$, and $v = \sin x$ with $v' = \cos x$. Then $u'v + uv' = \sin x + x\cos x$.

4 $\cos x \cdot e^{\sin x}$

Chain rule: the outer function is the exponential, which reproduces itself, and the inner derivative is $\cos x$. So the answer is $\cos x \cdot e^{\sin x}$.

5 $-\tan x$

Use $\dfrac{d}{dx}\ln f = \dfrac{f'}{f}$ with $f = \cos x$: the derivative is $\dfrac{-\sin x}{\cos x} = -\tan x$.

6 $\dfrac{1 - \ln x}{x^2}$

Quotient rule with $u = \ln x$ and $v = x$: $\dfrac{\frac{1}{x}\cdot x - \ln x \cdot 1}{x^2} = \dfrac{1 - \ln x}{x^2}$. It is zero at $x = e$, which is where $\dfrac{\ln x}{x}$ reaches its greatest value.

7 $\dfrac{x\sec^2 x - \tan x}{x^2}$

Quotient rule with $u = \tan x$ and $v = x$: $\dfrac{\sec^2 x \cdot x - \tan x \cdot 1}{x^2}$. Keep the order in the numerator — reversing it flips the sign.

8 $e^x(\sin x + \cos x)$

A product of two functions neither of which simplifies: $e^x \sin x + e^x \cos x$, and e^x factors out.

9 $\dfrac{2x}{x^2 + 1}$

Again $\dfrac{f'}{f}$, with $f = x^2 + 1$ and $f' = 2x$. The commonest error here is writing $\dfrac{1}{x^2 + 1}$ and dropping the inner derivative.

10 $3\sec^2(3x + 1)$

The outer derivative is $\sec^2(3x + 1)$ and the inner is 3, giving $3\sec^2(3x + 1)$.

Hard · hard

1 $\dfrac{1}{1 + \cos x}$

The quotient rule gives $\dfrac{\cos x(1 + \cos x) - \sin x(-\sin x)}{(1 + \cos x)^2} = \dfrac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2}$. Since $\cos^2 x + \sin^2 x = 1$ the numerator is $1 + \cos x$, which cancels one factor below, leaving $\dfrac{1}{1 + \cos x}$.

2 $\sec x$

Using $\dfrac{f'}{f}$ with $f = \sec x + \tan x$ and $f' = \sec x \tan x + \sec^2 x = \sec x(\tan x + \sec x)$. The bracket is f itself, so it cancels and the derivative is $\sec x$. This is why that logarithm is the antiderivative of $\sec x$.

3 $-\sin x \cdot \cos(\cos x)$

A chain with a trigonometric function inside another. The outer derivative is $\cos(\cos x)$ and the inner is $-\sin x$, giving $-\sin x \cdot \cos(\cos x)$. Keep the two arguments apart: the outer cosine is evaluated at $\cos x$.

4 $x^x(1 + \ln x)$

Neither the power rule nor the exponential rule applies, since both base and index vary. Take logarithms: $\ln y = x \ln x$. Differentiating, $\dfrac{y'}{y} = \ln x + x \cdot \dfrac{1}{x} = \ln x + 1$, so $y' = x^x(1 + \ln x)$.

5 $(\sin x)^x(\ln \sin x + x \cot x)$

Logarithmic differentiation again: $\ln y = x \ln(\sin x)$, so $\dfrac{y'}{y} = \ln(\sin x) + x \cdot \dfrac{\cos x}{\sin x} = \ln(\sin x) + x \cot x$. Multiplying back gives the answer.

6 $\dfrac{e^x}{(1 + e^x)^2}$

The quotient rule gives $\dfrac{e^x(1 + e^x) - e^x \cdot e^x}{(1 + e^x)^2}$. The numerator is $e^x + e^{2x} - e^{2x} = e^x$. The derivative is positive everywhere, so this function — the logistic curve — always increases.

7 $-\dfrac{1}{e}$, at $x = \dfrac{1}{e}$

$\dfrac{dy}{dx} = \ln x + 1$, zero when $\ln x = -1$, that is $x = \dfrac{1}{e}$. Then $\dfrac{d^2y}{dx^2} = \dfrac{1}{x} > 0$, so it is a minimum, of value $\dfrac{1}{e} \cdot (-1) = -\dfrac{1}{e} \approx -0.368$.

8 $-2e^{-x} \cos x$

First, $y' = -e^{-x} \sin x + e^{-x} \cos x = e^{-x}(\cos x - \sin x)$. Differentiating again, $y'' = -e^{-x}(\cos x - \sin x) + e^{-x}(-\sin x - \cos x) = e^{-x}(-\cos x + \sin x - \sin x - \cos x) = -2e^{-x} \cos x$.

9 Maximum $\sqrt{2}$ at $x = \dfrac{\pi}{4}$; minimum $-\sqrt{2}$ at $x = \dfrac{5\pi}{4}$

$\dfrac{dy}{dx} = \cos x - \sin x = 0$ gives $\tan x = 1$, so $x = \dfrac{\pi}{4}$ or $\dfrac{5\pi}{4}$. Then $\dfrac{d^2y}{dx^2} = -\sin x - \cos x$, which is $-\sqrt{2} < 0$ at $\dfrac{\pi}{4}$ and $\sqrt{2} > 0$ at $\dfrac{5\pi}{4}$. The values are $\pm\sqrt{2}$, which also follows from writing $y = \sqrt{2} \sin(x + \dfrac{\pi}{4})$.

10 $-\text{cosec}^2 x$

Write $\cot x = \dfrac{\cos x}{\sin x}$. The quotient rule gives $\dfrac{-\sin x \cdot \sin x - \cos x \cdot \cos x}{\sin^2 x} = \dfrac{-(\sin^2 x + \cos^2 x)}{\sin^2 x} = \dfrac{-1}{\sin^2 x} = -\text{cosec}^2 x$. The minus sign comes out of the numerator, not from anywhere else.